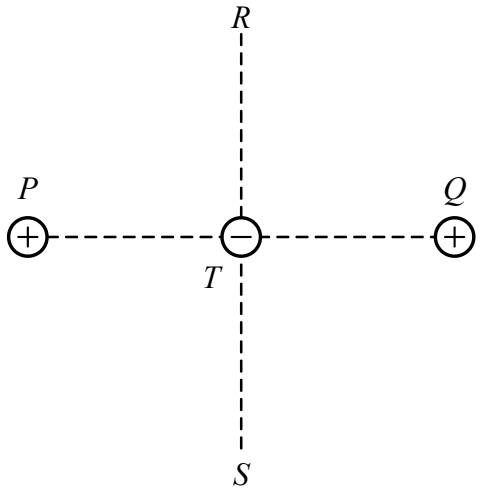
STPM 2017 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg, $N_A = 6.02 \times 10^{23}$ mol^{-1}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ $N\,m^2\,C^{-2}$, $\mu_0 = 4\pi \times 10^{-7}$ $H\,m^{-1}$.

Section A [15 marks]

1. Two identical positive point charges at P and Q are of equal distance from a negative point charge at T as shown in the diagram below.



When the charge at P is moved towards T, the negative charge will move towards

A P B Q C R D S

Answer: A

The attraction towards P increases as P comes closer, while the attraction towards Q is unchanged, so the negative charge moves towards P.

Explanation: Coulomb's force $\propto 1/r^2$.

2. A stationary lithium nucleus of charge $3e$ is bombarded by an energetic proton. What is the minimum speed of the proton in order to reach a distance of 1.0×10^{-15} m from the nucleus?

A 1.7×10^7 m s^{-1} B 2.9×10^7 m s^{-1} C 3.3×10^7 m s^{-1} D 5.0×10^7 m s^{-1}

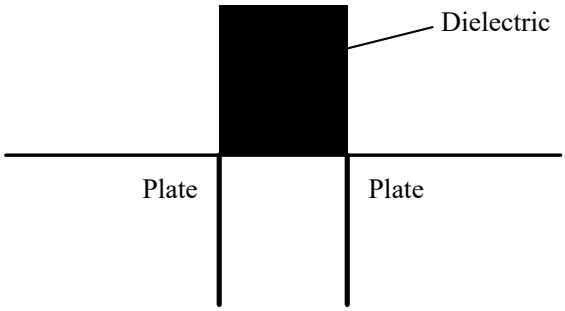
Answer: B

$$\frac{1}{2}mv^2 = \frac{k(3e)(e)}{r}$$

$$v = \sqrt{\left(\frac{2(9.0 \times 10^9)(3)(1.60 \times 10^{-19})^2}{(1.67 \times 10^{-27})(1.0 \times 10^{-15})}\right)} = 2.9 \times 10^7 \text{ m s}^{-1}$$

Explanation: All the kinetic energy becomes electric potential energy at the closest approach.

3. A dielectric is inserted into an air-filled parallel plate capacitor of capacitance 20 μF so that only half of the space between the plates is filled as shown in the diagram below.



If the dielectric constant is 5.0, what is the final capacitance of the capacitor?

A 8.3 μF B 30 μF C 60 μF D 100 μF

Answer: C

The dielectric fills half the plate area, so the capacitor acts as two halves in parallel:

$$C = \frac{20}{2} + 5.0 \times \frac{20}{2} = 10 + 50 = 60\ \mu\text{F}$$

Explanation: Side-by-side regions add like parallel capacitors.

4. A parallel-plate capacitor is initially charged by a battery and the battery is then disconnected. When the dielectric is removed from the charged capacitor,

A the charge stored in the capacitor increases
B the energy stored in the capacitor increases
C the potential difference between the plates decreases
D the electric field strength between the plates decreases

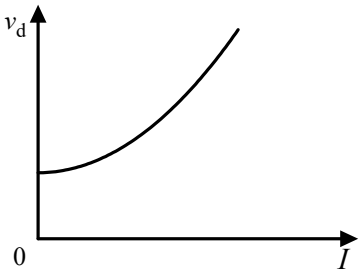
Answer: B

Q constant; removing the dielectric decreases C, so $U = \frac{Q^2}{2C}$ increases

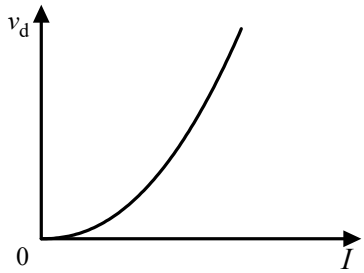
Explanation: Work must be done to pull the dielectric out; V and E increase.

5. Which graph shows the variation of drift velocity, v_d, of electrons with current, I, in a conductor?

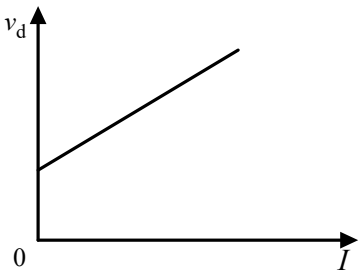
A



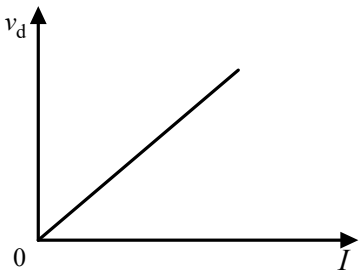
B



C



D



Answer: D

$I = nAev_d \Longrightarrow v_d \propto I$: a straight line through the origin (D)

Explanation: No current means no drift.

6. The conduction of electricity in metal is due to

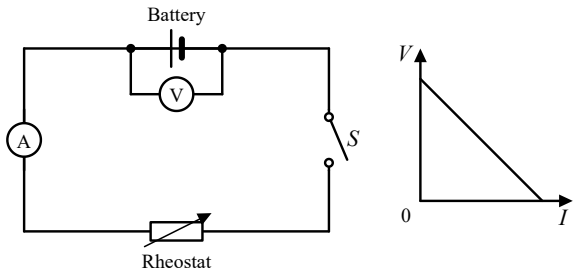
A the vibrations of ions in the metal
B the drifting of the free electrons in the metal
C the high random speed of the free electrons in the metal
D the collisions between the free electrons and the ions in the metal

Answer: B

Conduction is due to the drift of free electrons along the metal (B).

Explanation: The random motion alone carries no net charge.

7. A circuit used to determine the internal resistance of a battery is shown in the diagram below.



The variation of the voltmeter reading, V, with the ammeter reading, I, is shown in the graph above. The internal resistance of the battery can be determined from

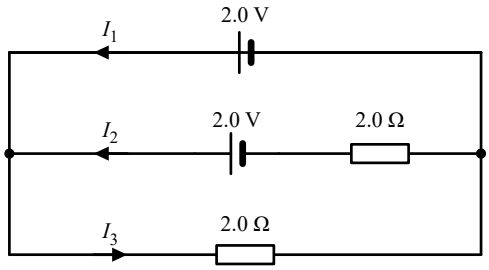
A the gradient of the graph
B the intercept on the I axis
C the intercept on the V axis
D the value of $\frac{V}{I}$ at any point on the graph

Answer: A

$V = \varepsilon - Ir$: the gradient of the V–I graph is $-r$

Explanation: The intercept on the V axis is the e.m.f.

8. Two identical batteries with each of e.m.f. 2.0 V and zero internal resistance, are connected to two resistors of resistance 2.0 Ω are shown in the diagram below.



What are the values of the currents I_1, I_2 and I_3?

	I_1	I_2	I_3
A	0 A	1.0 A	1.0 A
B	0.5 A	0.5 A	1.0 A
C	1.0 A	0 A	1.0 A
D	2.0 A	0 A	2.0 A

Answer: C

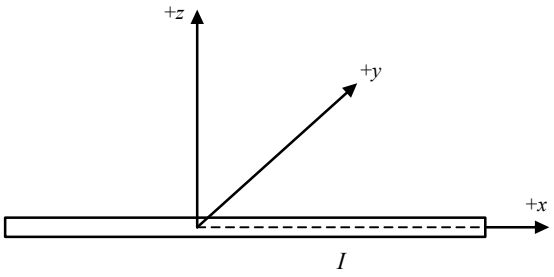
The top cell (no internal resistance) holds the two junctions 2.0 V apart.

Middle branch: its 2.0 V cell exactly opposes this p.d., so $I_2 = 0$.

Bottom branch: $I_3 = \frac{2.0}{2.0} = 1.0$ A, so $I_1 = 1.0$ A

Explanation: Check with Kirchhoff's first law: $I_1 = I_2 + I_3$.

9. A current, I, flows through a conductor in the direction of $+x$ is shown in the diagram below.



The magnetic force, F, per unit length experienced by the conductor is 0.20 N m^{-1} in the direction of $+y$. If the current, I, is 10.0 A, which statement is **not** true about the magnetic field, B?

A The magnitude of B is 0.02 T.
B The direction of B is in the $-z$ axis.
C The direction of B is in the $+z$ axis.
D The direction of B is perpendicular to I and F.

Answer: C

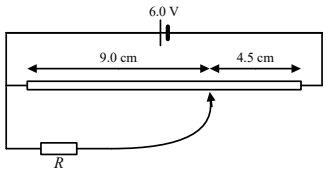
$$B = \frac{F/l}{I} = \frac{0.20}{10.0} = 0.020 \text{ T}$$

$\mathbf{F} = \mathit{Il} \times \mathbf{B}$: $+x \times (-z) = +y$, so **B** is along $-z$. Statement C ($+z$) is not true.

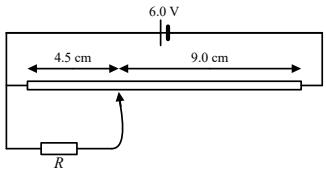
Explanation: B must be perpendicular to both *I* and *F*.

10. Which circuit shows the potential divider that provides potential difference of 2.4 V to the load of resistance, *R*?

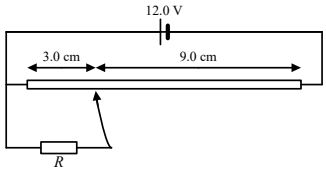
A



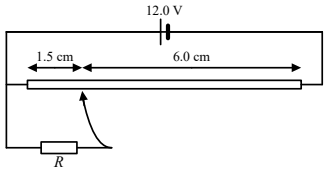
B



C



D



Answer: D

The load takes the p.d. across the part of the wire between the end it is joined to and the slider.

$$\text{D: } 12.0 \times \frac{1.5}{1.5 + 6.0} = 2.4 \text{ V}$$

Explanation: A gives 4.0 V, B gives 2.0 V, C gives 3.0 V (assuming the load draws little current).

11. A small surface area is placed inside a uniform magnetic field. The surface is inclined at an angle, θ, which is the angle between the magnetic field and normal to the plane of the surface. Which statement is **not** true about the magnetic flux through the surface?

A When θ increases, the magnetic flux through the surface decreases.
B When θ decreases, the magnetic flux through the surface increases.
C The magnetic flux through the surface is independent of θ.
D The magnetic flux through the surface is maximum when $\theta = 0°$.

Answer: C

$\Phi = BA \cos \theta$: the flux depends on θ, so statement C is not true

Explanation: Flux is maximum at $\theta = 0°$ and decreases as θ increases.

12. An inductor of length 0.20 m consists of 200 turns and cross-sectional area of 2.5×10^{-4} m^2. If a current of 1.2 A flows in the inductor, what is the energy stored in the inductor?

A 1.8×10^{-6} J B 4.5×10^{-5} J C 9.0×10^{-5} J D 2.3×10^{-4} J

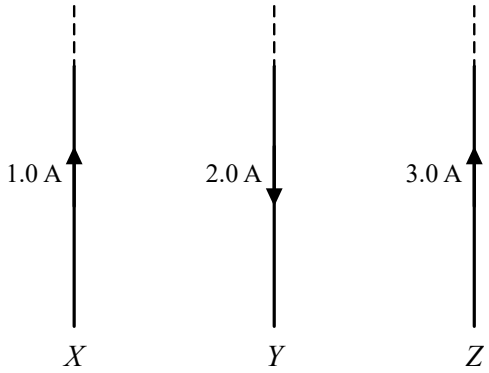
Answer: B

$$L = \frac{\mu_0 N^2 A}{l} = \frac{(4\pi \times 10^{-7})(200)^2(2.5 \times 10^{-4})}{0.20} = 6.28 \times 10^{-5}\ \text{H}$$

$$U = \frac{1}{2}LI^2 = \frac{1}{2}(6.28 \times 10^{-5})(1.2)^2 = 4.5 \times 10^{-5}\ \text{J}$$

Explanation: Find L from the dimensions first.

13. Three similar long thin wires, X, Y and Z which are parallel to each other are shown in the diagram below.



The current in X, Y and Z are 1.0 A, 2.0 A and 3.0 A respectively. If Y is placed 20 cm from both X and Z, what are the magnitude and direction of the resultant force per unit length on Y?

	Magnitude	*Direction*
A	4.0×10^{-6} N m^{-1}	To the left
B	4.0×10^{-6} N m^{-1}	To the right
C	8.0×10^{-6} N m^{-1}	To the left
D	8.0×10^{-6} N m^{-1}	To the right

Answer: A

Y carries current opposite to both X and Z, so both repel it.

From X (to the right): $\frac{\mu_0(1.0)(2.0)}{2\pi(0.20)} = 2.0 \times 10^{-6}$ N m^{-1}; from Z (to the left): $\frac{\mu_0(2.0)(3.0)}{2\pi(0.20)} = 6.0 \times 10^{-6}$ N m^{-1}

Resultant = 4.0×10^{-6} N m^{-1} to the left.

Explanation: Unlike currents repel.

14. A pure inductor is connected to an alternating current supply of constant peak voltage but of frequency which can be varied. Which statement is true about the peak current in the circuit when the frequency is increased?

A The peak current decreases and lags the peak voltage more than $\frac{\pi}{2}$ radians.

B The peak current increases and lags the peak voltage less than $\frac{\pi}{2}$ radians.

C The peak current increases and lags the peak voltage by $\frac{\pi}{2}$ radians.

D The peak current decreases and lags the peak voltage by $\frac{\pi}{2}$ radians.

Answer: D

$I_0 = \frac{V_0}{2\pi fL}$: the peak current decreases as f increases, and it always lags the voltage by $\frac{\pi}{2}$ rad

Explanation: The phase difference in a pure inductor does not depend on frequency.

15. A sinusoidal current, I, is given by $I = 12 \sin 200\pi t$, where I is in amperes and t in seconds. If the sinusoidal current flows through a capacitor of capacitance 3300 μF, what is the r.m.s. voltage across the capacitor?

A 4.1 V B 5.8 V C 18 V D 25 V

Answer: A

$$I_{rms} = \frac{12}{\sqrt{2}} = 8.49 \text{ A}; X_C = \frac{1}{(200\pi)(3300 \times 10^{-6})} = 0.482 \ \Omega$$

$$V_{rms} = I_{rms} X_C = (8.49)(0.482) = 4.1 \text{ V}$$

Explanation: $\omega = 200\pi$ rad s^{-1} is read from the equation.

Section B [15 marks]

16. (a) Explain the motion of a charged particle moving perpendicularly in a

(i) uniform electric field, [2]

Answer

The particle experiences a constant force qE perpendicular to its initial velocity (constant acceleration)	1
Its velocity along the original direction is unchanged, so it follows a parabolic path	1
	2

Explanation: Like a horizontally projected object under gravity.

(ii) uniform magnetic field. [2]

Answer

The magnetic force qvB is always perpendicular to the velocity and does no work — 1

The speed is constant and the particle moves in a circle — 1

2

Explanation: $r = mv/qB$.

(b) An electron of mass, m, is accelerated by a potential difference, V, in a region of electric field, E, and magnetic field, B, perpendicular to each other. The electron travels in a straight line. Derive an expression for the ratio $\frac{e}{m}$. [4]

Answer

Straight line: electric force = magnetic force, $eE = evB$ — 1

$v = \frac{E}{B}$ — 1

Accelerated through V: $eV = \frac{1}{2}mv^2 \Longrightarrow \frac{e}{m} = \frac{v^2}{2V}$ — 1

$\frac{e}{m} = \frac{E^2}{2VB^2}$ — 1

4

Total for question **8**

Explanation: This is the basis of Thomson's e/m experiment.

17. In a dynamic microphone, sound waves enter the microphone and vibrate a diaphragm that houses an induction coil in a magnetic field. The energy of sound is 0.350 J and all the energy is transferred to the diaphragm within 0.800 s. The resistance of the coil is 4.00 Ω.

(a) State the transformation of energy in the dynamic microphone. [1]

Answer

Sound energy → kinetic energy of the diaphragm and coil → electrical energy — 1

1

Explanation: The moving coil cuts the magnetic field lines.

(b) Determine the average induced e.m.f. in the coil. [4]

Answer

Power = $\frac{0.350}{0.800}$ = 0.4375 W — 1

$P = \frac{\varepsilon^2}{R}$ — 1

$\varepsilon = \sqrt{((0.4375)(4.00))}$ — 1

ε = 1.32 V — 1

4

Explanation: All the sound energy is converted to electrical energy in the coil.

(c) Determine the change of magnetic flux through the coil. [2]

Answer

$\varepsilon = \frac{\Delta(N\Phi)}{\Delta t} \Longrightarrow \Delta(N\Phi) = (1.32)(0.800)$ — 1

= 1.06 Wb — 1

2

Total for question **7**

Explanation: This is the change in flux linkage of the coil.

Section C [30 marks]

18. (a) State *electric potential at a point* in an electric field. [1]

Answer

The work done per unit positive charge in bringing a small positive test charge from infinity to that point — 1

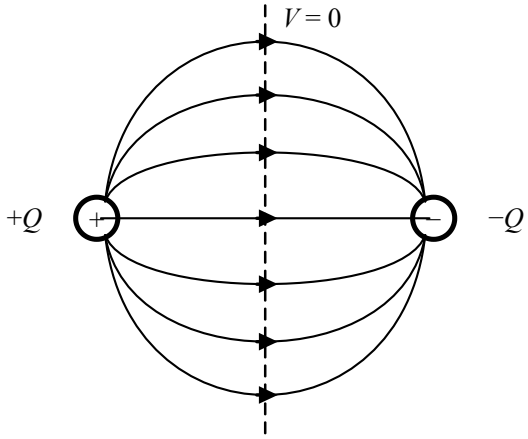
1

Explanation: $V = W/q$; unit volt (J C^{-1}).

(b) Two point charges $+Q$ and $-Q$ are separated at a distance, x, from each other.

(i) Draw a diagram to show the electric field lines produced by the charges. [2]

Answer



Field lines leave $+Q$ and end on $-Q$	1
Arrows from + to −; lines never cross	1
	2

Explanation: The pattern is symmetrical about the line joining the charges.

(ii) In your diagram, draw an equipotential line where the electric potential is zero. [1]

Answer

The zero equipotential is the perpendicular bisector of the line joining the charges (dashed line)	1
	1

Explanation: Every point on it is the same distance from $+Q$ and $-Q$.

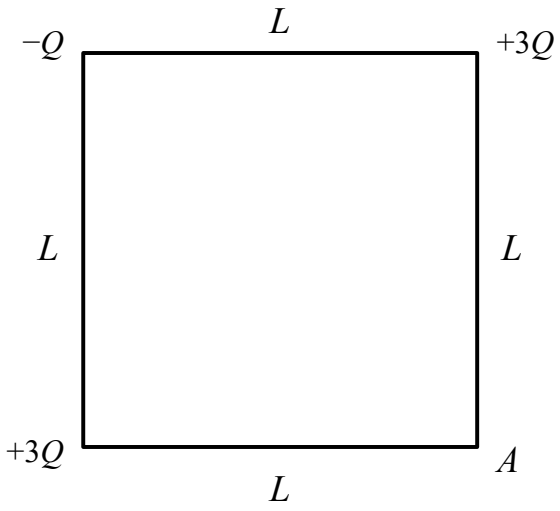
(iii) Derive an expression for the work done to separate the charges to infinity. [5]

Answer

Potential at $-Q$ due to $+Q$: $V = \frac{Q}{4\pi\varepsilon_0 x}$	1
Initial potential energy $U = (-Q)V = -\frac{Q^2}{4\pi\varepsilon_0 x}$	1
At infinite separation the potential energy is zero	1
Work done = $U_{\text{final}} - U_{\text{initial}} = 0 - (-\frac{Q^2}{4\pi\varepsilon_0 x})$	1
$W = \frac{Q^2}{4\pi\varepsilon_0 x}$	1
	5

Explanation: Positive work must be done against the attraction.

(c) Three point charges $-Q$, $+3Q$ and $+3Q$ are placed at the corners of a square dimension, L, as shown in the diagram below.



In terms of Q, L and ε_o, determine an expression for the

(i) electric field strength at point A, [4]

Answer

Each $+3Q$ (distance L): $E = \frac{3Q}{4\pi\varepsilon_0 L^2}$, one along each side, at 90°; resultant $\frac{3\sqrt{2}Q}{4\pi\varepsilon_0 L^2}$ along the diagonal away from $-Q$ — 1

$-Q$ (distance $\sqrt{2}L$): $E = \frac{Q}{4\pi\varepsilon_0(2L^2)}$ along the diagonal towards $-Q$ — 1

Net $E = \frac{Q}{4\pi\varepsilon_0 L^2}(3\sqrt{2} - \frac{1}{2}) = \frac{3.74Q}{4\pi\varepsilon_0 L^2} = \frac{0.30Q}{\varepsilon_0 L^2}$ — 1

Directed along the diagonal, away from $-Q$ — 1

4

Explanation: All three contributions lie along (or combine to lie along) the diagonal through A.

(ii) electric potential at point A. [2]

Answer

$V = \frac{1}{4\pi\varepsilon_0}(\frac{3Q}{L} + \frac{3Q}{L} - \frac{Q}{\sqrt{2}L})$ — 1

$V = \frac{5.29Q}{4\pi\varepsilon_0 L} = \frac{0.42Q}{\varepsilon_0 L}$ — 1

2

Total for question **15**

Explanation: Potential is a scalar: add with signs, no directions.

19. (a) Define *drift velocity*. [1]

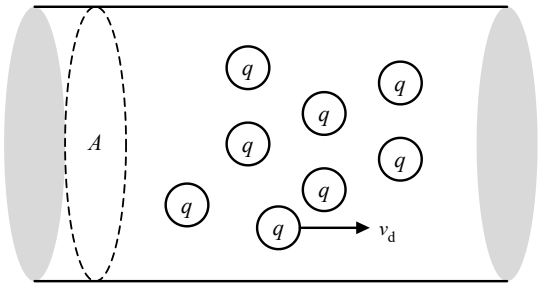
Answer

The average velocity of the free charge carriers along a conductor when an electric field (p.d.) is applied — 1

1

Explanation: It is much smaller than their random speed.

(b) Charge carries with each of charge, q, moving with drift velocity, v_d, in a segment of wire of cross-sectional area, A, is shown in the diagram below.



Derive an expression for the relationship between current in the wire and drift velocity. [4]

Answer

In time t each carrier moves a distance $v_d t$; the volume passing a cross-section is $Av_d t$ — 1

Number of carriers in this volume = $nAv_d t$ — 1

Charge passing = $nAv_d tq$ — 1

$I = \frac{Q}{t} = nAqv_d$ — 1

4

Explanation: n is the number of carriers per unit volume.

(c) Household wiring contains copper wire which has a diameter of 2.053 mm and the maximum current allowed is 20.0 A.

[The density of copper is 8.80×10^3 kg m^{-3}, molar mass of copper is 63.54 g mol^{-1}.]

(i) Calculate the density of free electrons in copper if each copper atom contributes one free electron. [4]

Answer

Moles per m^3 $= \dfrac{8.80 \times 10^3}{63.54 \times 10^{-3}} = 1.385 \times 10^5$ mol m^{-3} — 1

Atoms per m^3 $= (1.385 \times 10^5)(6.02 \times 10^{23})$ — 1

One free electron per atom — 1

$n = 8.34 \times 10^{28}$ m^{-3} — 1

4

Explanation: Convert g mol^{-1} to kg mol^{-1}.

(ii) Calculate the current density. [3]

Answer

$A = \pi(1.0265 \times 10^{-3})^2 = 3.31 \times 10^{-6}$ m^2 — 1

$J = \dfrac{I}{A} = \dfrac{20.0}{3.31 \times 10^{-6}}$ — 1

$J = 6.04 \times 10^6$ A m^{-2} — 1

3

Explanation: Use the radius, half of 2.053 mm.

(iii) Calculate the drift velocity of electrons in the wire. [1]

Answer

$v_d = \dfrac{J}{ne} = \dfrac{6.04 \times 10^6}{(8.34 \times 10^{28})(1.60 \times 10^{-19})} = 4.53 \times 10^{-4}$ m s^{-1} — 1

1

Explanation: About half a millimetre per second.

(d) Explain the effect of an increase in temperature on the drift velocity of the electrons in the copper wire. [2]

Answer

A higher temperature makes the ions vibrate more, so collisions are more frequent (shorter mean free time) and the resistance increases — 1

For the same p.d. the current falls, so the drift velocity decreases — 1

2

Total for question **15**

Explanation: $v_d = eE\tau/m$: a smaller τ gives a smaller drift velocity.

20. (a) Define *root mean square current of an alternating current*. [1]

Answer

The value of the steady (direct) current that produces the same heating effect (power) in a resistor as the alternating current — 1

1

Explanation: For a sinusoidal current $I_{rms} = I_0/\sqrt{2}$.

(b) A filament bulb is connected in series with a sinusoidal voltage supply. Explain how the brightness of the bulb changes if the frequency of the supply is increased but its root mean square voltage remains constant. [2]

Answer

The resistance of the filament does not depend on frequency — 1

so the r.m.s. current and power are unchanged: the brightness stays the same — 1

2

Explanation: Only reactances depend on frequency.

(c) A filament bulb labelled "100 V, 40 W" is connected in series with a resistor and sinusoidal voltage supply of r.m.s voltage 240 V and frequency 50 Hz. If the bulb lit with a normal brightness, calculate

(i) the resistance of the bulb, [2]

Answer

$R = \frac{V^2}{P} = \frac{100^2}{40}$ — 1

$R = 250\ \Omega$ — 1

2

Explanation: Use the bulb's rated values.

(ii) the resistance of the resistor. [3]

Answer

$I = \frac{40}{100} = 0.40$ A — 1

P.d. across the resistor = 240 − 100 = 140 V — 1

$R = \frac{140}{0.40} = 350\ \Omega$ — 1

3

Explanation: Resistor and bulb are in phase, so their p.d.s add directly.

(d) The filament bulb in (c) above is then connected in series with a capacitor and the same voltage supply. If the bulb lit with a normal brightness, calculate

(i) the capacitance of the capacitor, [4]

Answer

$V_C = \sqrt{(240^2 - 100^2)} = 218$ V — 1

$X_C = \frac{218}{0.40} = 545\ \Omega$ — 1

$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(50)(545)}$ — 1

$C = 5.84 \times 10^{-6}$ F — 1

4

Explanation: The capacitor and bulb p.d.s are 90° out of phase, so they add as phasors.

(ii) the phase angle between the current and voltage supply. [2]

Answer

$\cos \varphi = \frac{V_R}{V} = \frac{100}{240}$ — 1

$\varphi = 65.4°$, with the current leading the supply voltage — 1

2

Explanation: Equivalently $\tan \varphi = X_C/R = 545/250$.

(e) State **one** advantage of using capacitors instead of resistors to regulate current in an alternating current circuit. [1]

Answer

A capacitor dissipates no power (no energy wasted as heat) 1

1

Total for question **15**

Explanation: Its average power over a cycle is zero.